

AI PRIVATE TUTOR

Personalized Telegram-Integrated LMS & Student Helper Platform



Web Portal: <https://senangbelajar.web.id>



Telegram Bot: [@senangbelajar_bot](https://t.me/senangbelajar_bot)

Platform Architecture & Tech Stack

✓ Next.js 14 (App Router)

Powers the main responsive student dashboard & admin controls.

✓ Telegraf.js Webhooks

Handles instant dual-channel interactions via @senangbelajar_bot.

✓ Prisma & PostgreSQL

Provides reliable, type-safe database schemas with migration tracking.

✓ SumoPod LLM Gateway

Runs deepseek-v4-flash for cost-efficient AI private tutoring.

Telegram Client (Student Chat)

Next.js backend (Webhook /api/bot)

DB: PostgreSQL | ORM: Prisma

SumoPod LLM Gateway (deepseek-v4-flash)

Core Platform Features

Dual Curriculum Integration

Unifies Kemendikdasmen's SIBI (national textbooks) and school-specific Moodle portals into a single learning path.

YouTube Video Linker

Scrapes, filters, and maps high-quality reference videos to curriculum topics. Validated via oEmbed API.

Adaptive Quiz Engine

Generates and reviews student knowledge using dynamic question banks generated directly matching progress.

React Flow Mindmaps

Automatically visualizes curriculum outlines and study guides into interactive node maps on the student dashboard.

Operations & Default Schedule



Default Study Schedule

- **Intensive Schedule:** Senin, Rabu, Jumat

19:00 - 21:00 WIB (UTC+7)

- **Reguler Schedule:** Selasa, Kamis, Minggu

19:00 - 20:00 WIB (UTC+7)

- **Notification Policy:** Reminder sent exactly 15 mins before (18:45 WIB) with Topic and Sub-topic selection.



Moodle Automation Pipeline

Automated scrapers fetch Moodle PDFs, metadata, and lesson structures securely using user enrolment tokens.



Discrepancy Checking Engine

Periodic background jobs check physical directories against database tables to match student-specific courses.



Robust Hosting Infrastructure

Managed PM2 application runners and Caddy server with SSL handling incoming API calls and bot webhooks.